

Topic Modelling for Dark Web Monitoring

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Abstract

This dissertation explores topic modeling techniques to enhance the monitoring of Dark Web forums, focusing on detecting rare yet critical terms often overlooked in general discussions. Five models were investigated: Baseline Latent Dirichlet Allocation (LDA), LDA with Named Entity Recognition (NER) and nouns as well as proper nouns, Modified LDA, Non-Negative Matrix Factorization (NMF), and BERTopic. A novel Modified LDA model was introduced, incorporating dynamic word influence to prioritize rare, domain-specific terms within the topic-word distributions.

The models were evaluated using two primary metrics, Topic Diversity, which measures the uniqueness of terms across topics, and Inverse Word Embedding Coherence (IWEC) scores, assessing the semantic relationships between top words within topics. The Modified LDA model successfully captured exclusive, low-frequency terms critical to Dark Web discourse that were missed by traditional models, demonstrating its capability to highlight contextually significant yet sparse vocabulary. By applying these techniques yearly, the models provided valuable insights into the evolving trends and emerging threats within Dark Web forums.

This research contributes to the field by refining topic modeling approaches to detect and emphasize low-frequency, high-impact terms, thereby enhancing the effectiveness of Dark Web monitoring.

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1 Introduction

1.1 Overview Of Dark Web Monitoring.

The Dark Web is that part of the internet hidden from public view, only accessible through encrypted networks like Tor or I2P. These networks add anonymity to users by routing the communications they make through so many servers that their identity and activity are effectively masked. These networks were originally offered as an effort to enable people to have more privacy in their actions and ensure freedom of speech but have been highly abused for illegal activities ranging from drug trafficking to cybercrime and the trade of stolen data (Biryukov et al. (2013)).

At the core of the Dark Web are forums and marketplaces, which serve as hubs for interaction and commerce. Forums, resembling traditional message boards, are where users discuss a range of illicit activities, such as hacking techniques, identity theft, or how to exploit vulnerabilities in software. These spaces are critical for knowledge exchange, with users building reputations based on the quality of the information they provide. Marketplaces, meanwhile, allow users to buy and sell goods such as counterfeit documents, stolen credit card details, and malware. These transactions are typically carried out using cryptocurrencies like Bitcoin, ensuring anonymity for both parties (Alharbi et al. (2021)). The emphasis on trusted knowledge exchange and reputation building generates a rich source of text data.

Despite the text-rich nature of Dark Web forums and marketplaces, monitoring these spaces is highly challenging due to their fluid and secretive nature. Conveyed messages are mostly in coded language with continuously changing slang deliberately introduced to not be caught. Besides, a large size of activity on forums and marketplaces would complicate manual tracking of trends. Traditional keyword-based searches are typically not effective because users adapt their language to avoid detection, so the detection and understanding of new terms is always being introduced to mask illicit activities (Tavabi et al. (2019)). Being able to detect and understand these shifts in language and behavior is important in realizing emerging threats.

This is where **Natural Language Processing** (NLP) becomes an essential tool. NLP allows machines to process and analyze human language in a way that captures not only the meaning of individual words but also the relationships between them within a broader context. This is particularly important on the Dark Web, where discussions can be cryptic and constantly changing. Instead of relying on predefined keywords, NLP can identify patterns and topics based on how words are used together, enabling a more dynamic and adaptable approach to monitoring (Hughes et al. (2020)).

Topic modeling is one of the most effective methods in natural language processing we could use for Dark Web surveillance. By spotting groups of words that often coexist, topic modeling automatically finds the major themes in a body of text. These clusters of words form “*topics*” that give insight into what is being discussed, without the need for predefined categories or terms. This is especially useful in environments like the Dark Web, where the language is informal, unstructured, and constantly shifting (Blei et al. (2003)).

For example, in a forum dedicated to hacking, topic modeling might reveal a growing discussion around a new type of malware, even if the exact name of the malware is not consistently used. By identifying clusters of words associated with certain hacking techniques or vulnerabilities, topic modeling can flag new areas of interest and potentially uncover emerging threats before they spread widely. This allows for monitoring of trends as they develop, providing critical insights that would be difficult to obtain through manual analysis or basic keyword searches (Ursani et al. (2021)).

1.2 Problem Definition

A key challenge in Dark Web monitoring is the detection of critical, low-frequency terms that are often sparse and overlooked in general discussions. These terms may point to significant activities by specific groups or not but are frequently dismissed by broader communities as scams or hoaxes due to their infrequent appearance. This assumption that such discussions are irrelevant leads to the neglect of potentially important topics, hindering effective monitoring and threat detection (Nazah et al. (2020)).

The difficulty lies in the fact that these terms appear infrequently, making them hard to detect using traditional frequency-based models. Standard approaches focus on high-frequency or widely discussed terms, which means that less common but potentially crucial terms remain hidden amidst general chatter. Moreover, even frequently occurring important terms can be overshadowed by irrelevant or non-contextual content, making it difficult to isolate the most meaningful information (Hofstätter et al. (2019)).

Given the unstructured and ambiguous nature of Dark Web communications, these sparse yet significant discussions can easily be missed. This problem is further compounded by the evolving and adaptive language used to avoid detection, requiring more sophisticated methods to extract relevant information.

The problem addressed by this research is the need for a novel topic modelling framework capable of identifying and prioritizing both low frequency but contextually significant terms, as well as frequently occurring critical terms that may be buried in general discussions. By addressing the limitations of existing approaches, this research seeks to improve the ability to detect subtle yet important trends on the Dark Web, offering a more nuanced and effective approach to monitoring.

1.3 Research Objectives

This study aims to further develop and enhance existing topic modeling techniques to better capture critical but often infrequently mentioned terms within Dark Web discussions. The focus is on modifying current models to prioritize the detection of low-frequency yet highly significant terms, which, although sparse, are essential for identifying key activities that broader communities often dismiss as irrelevant or hoaxes. Despite their infrequency, these terms can provide valuable insights, and the goal is to ensure they are accurately identified and not overlooked.

The research also seeks to improve the recognition of frequently occurring yet contextually important terms, which can easily get lost in general discussions and irrelevant content. By refining the models to better differentiate between noise and crucial domain-specific terms, the aim is to enhance the system's ability to extract meaningful insights from the vast amount of general chatter.

Additionally, the research will further integrate Named Entity Recognition (NER) to entities within the unstructured, dark language typical of Dark Web communications. This will provide a deeper understanding of the what entities are involved in illicit activities, addressing a challenge that existing models struggle to handle effectively.

Finally, the study will focus on optimizing topic coherence and relevance by refining the models to generate more meaningful and contextually appropriate topics. These improvements will be achieved by integrating noun phrases and modifying existing models to emphasize detecting important but underrepresented terms.

1.4 Research Questions & Hypothesis

1.4.1 Research Questions:

- RQ1: What are the main topics discussed in Dark Web forums over a specific period?
- RQ2: What machine learning and natural language processing techniques are most effective for detecting and analysing topics in Dark Web forums?
- RQ3: Can we develop techniques to discover exclusive rare words in topic distributions in Dark Web forums?

1.4.2 Hypotheses:

- H1: Traditional topic modelling techniques such as Latent Dirichlet Allocation (LDA) will effectively identify main topics discussed in Dark Web forums.
- H2: Techniques combining word embeddings will demonstrate enhanced capability in capturing semantic nuances in Dark Web conversations.
- H3: Enhancing traditional topic models with named entity recognition (NER) and nouns more will lead the discovery of exclusive and domain-specific words.

Above listed Research questions and hypotheses primarily follow previously submitted research proposal for this project

1.5 Ethical and Legal Considerations

The data for this research was sourced from cafedread.net, a clear web archive of public discussions from the Dread forum on the Dark Web. Ethical standards were maintained throughout, with no personally identifiable information (PII) included in any part of the project. The data was accessed in a read-only format to ensure no interaction with hidden services or encrypted content. Forums or discussions that posed ethical or legal concerns were deliberately excluded to keep the research focused on relevant topics.

Legal Compliance: This research adhered to legal standards, including GDPR. Data was sourced legally from cafedread.net before the website's current status affected compliance. All scraping processes followed the site's terms of service, and the dataset was securely stored in encrypted environments with restricted access.

Risk Mitigation: Data collection was limited to publicly available content, avoiding sensitive or illegal material. No engagement with forums related to illegal activities or highly sensitive content occurred. Data was securely encrypted, and access was strictly controlled to ensure confidentiality.

2 Literature Review

Topic modeling is a common NLP application that uses statistical methods to uncover hidden themes in large text corpora, mirroring how humans identify recurring words and contextual themes to categorize content. However, this automation often oversimplifies nuanced human understanding, potentially missing contextual subtleties.

Despite advancements aiming to address inherent challenges, topic models haven't fully captured the complexity of human language. This literature review critically examines foundational models, acknowledging their contributions and limitations, and evaluates how they've

influenced subsequent developments often perpetuating existing issues rather than resolving them.

Non-Negative Matrix Factorization (NMF), an early algorithm introduced by [Lee & Seung \(1999\)](#), was later applied to topic modeling in text analysis ([Xu et al. 2003](#)). According to [Berry et al. \(2007\)](#), NMF divides a non-negative document-term matrix into two non-negative matrices, one representing topics and the other representing document associations with these topics. This method uses an intuitive linear algebraic framework to express each document as a combination of latent topics with word distributions. NMF struggles with uncertainty in topic assignments due to its lack of probabilistic foundation ([Blei et al. 2003](#)).

Probabilistic Latent Semantic Indexing (pLSI), presented by [Hofmann \(1999\)](#), was an initial probabilistic method for topic modeling. pLSI models each word in a document as a combination of topics, where each topic represents a probability distribution over words. In contrast to NMF, pLSI provides a probabilistic framework in which each document is linked to a mixture of topics, and each topic is defined by a distribution over words ([Griffiths 2004](#)). Nonetheless, pLSI possesses a considerable limitation as it fails to offer a generative model for documents beyond the training set, as it lacks a prior distribution over topics for unobserved documents. This leads to overfitting and restricts its generalizability ([Steyvers & Griffiths 2007](#)).

Latent Dirichlet Allocation (LDA), introduced by [Blei et al. \(2003\)](#), is a foundational method that provided a new perspective on topic modeling. LDA offers a probabilistic generative model for documents where every topic is seen as a distribution over words and each document is considered as a mixture of topics. This probabilistic framework allows LDA to handle uncertainty more effectively and provides a theoretical basis for inferring hidden topic structures from observed data. LDA aims to infer the hidden topic structure by estimating the posterior distribution of the topics given the words in the documents. This involves maximizing the likelihood of the observed corpus under the LDA model, which is achieved using variational inference or Gibbs sampling due to the intractability of exact inference. Since its inception, LDA has been widely for real world applications and is still used for baselines. However it has notable drawbacks, it requires specifying the number of topics in advance relies on a bag-of-words approach (which ignores word order and context) and lacks the ability to capture contextual nuances.

To overcome these limitations and adapt LDA to various domains, researchers have developed various extensions that enhance the original framework's flexibility, scalability, and capacity to model complex textual relationships notable extensions have been stated.

1. *Hierarchical Dirichlet Process (HDP)* ([Yee Whye Teh & Blei 2006](#)) addresses LDA's constraint of needing a predefined number of topics. HDP employs a nonparametric Bayesian approach using Dirichlet Processes, allowing the number of topics to be inferred directly from the data. It constructs an infinite mixture model in which the number of topics can dynamically grow with the complexity of the dataset. HDP introduces a hierarchical structure with a global Dirichlet Process defining a shared set of topics across the corpus, while each document draws its topic distribution from a local Dirichlet Process. Despite its flexibility, HDP still adheres to the bag-of-words assumption and can be computationally intensive due to its iterative nature ([Teh & Jordan 2010](#)).
2. *Correlated Topic Model (CTM)* ([Lafferty & Blei 2005](#)) extends LDA by addressing the independence assumption between topics. In LDA, topics are assumed to be uncorrelated, which often does not reflect the true nature of language where topics are interdependent. CTM introduces a logistic normal distribution in place of the Dirichlet prior over document-topic distributions, allowing the model to capture correlations among topics. This approach provides a more nuanced thematic representation by modeling how topics naturally co-occur within documents. However, CTM, like LDA, is constrained by the bag-of-words assumption and does not incorporate temporal dynamics.

3. *Dynamic Topic Model (DTM)* (Blei & Lafferty 2006) further extends LDA to account for the evolution of topics over time. By introducing a sequential framework that links topic distributions across different time slices, DTM models how topics shift and change throughout a temporal sequence. This makes it particularly suitable for analyzing time-stamped documents, revealing insights into how thematic content evolves. Nevertheless, DTM inherits certain limitations from LDA, such as ignoring word order and context within individual time slices.
4. *Online LDA* (Hoffman et al. 2010) modifies LDA to be more scalable for large-scale and streaming datasets. Standard LDA requires processing the entire corpus in a batch, which can be computationally prohibitive for large datasets. Online LDA employs stochastic variational inference, enabling the model to process data in mini-batches and update incrementally. This approach enhances LDA's applicability to real-time data analysis and large corpora, making it more efficient while maintaining the probabilistic framework of LDA.
5. *Biterm Topic Model (BTM)* (Yan et al. 2013) is designed to improve topic modeling for short texts, where traditional LDA often struggles due to sparse word co-occurrences. BTM directly models word co-occurrences (biterms) across the entire corpus, rather than within individual documents. By focusing on capturing the word pairings that occur throughout the dataset, BTM enhances topic coherence in short texts, such as social media posts and news headlines, where data is often sparse and fragmented.

While these extensions addressed key issues, they still relied on the bag-of-words assumption, missing the semantic richness captured by word embeddings. Word2vec (Mikolov 2013) introduced static embeddings that capture semantic and syntactic word relationships by embedding words in a continuous vector space. However, these static embeddings assign a single vector per word, failing to capture contextual variations in word meanings (Pennington et al. 2014).

To address this, LDA2Vec (Moody 2016) integrates word embeddings with LDA, combining topic models with neural embeddings. LDA2Vec jointly learns word vectors and Dirichlet-distributed document-topic mixtures, capturing both global thematic structures and local word semantics. However, it inherits limitations from static embeddings, such as ignoring word sense variations (Camacho-Collados & Pilehvar 2018), and requires more computational resources due to the joint learning of word and topic vectors. Nonetheless, LDA2Vec represents a significant step forward by bridging probabilistic topic models with richer semantic representations.

Despite these advancements, topic modeling still faced challenges in capturing complex document-topic distributions and high-dimensional text data. To address these challenges, researchers began exploring the integration of deep learning architectures into topic modeling. This led to the development of Neural Topic Models (NTMs), which leverage the power of neural networks to uncover more intricate patterns in text data.

Neural Topic Models (NTMs) utilize deep learning frameworks to identify complex patterns in text that traditional approaches such as LDA frequently overlook (Bengio et al. 2013). The Neural Variational Document Model (NVDM) (Miao et al. 2016) utilizes Variational Autoencoders (VAEs) (Kingma 2013) to project documents into a continuous latent space. Each document d is defined by a latent variable z , drawn from a Gaussian distribution, which is subsequently processed through a neural network to reconstruct the document's word distribution using a softmax function (Miao et al. 2016). NVDM optimizes this process by maximizing the Evidence Lower Bound (ELBO) via variational inference (David M. Blei & McAuliffe 2017), enabling it to handle high-dimensional data and capture complex document-topic relationships.

This approach allows NTMs to better model the semantic richness and nuances in natural language, offering advantages in terms of flexibility and scalability over traditional topic models (Dieng et al. 2020).

ProdLDA Srivastava & Sutton (2017) builds on this by using a 'product of experts' framework (Hinton 2002). Instead of a single latent variable, ProdLDA models topics as the product of multiple latent variables, enhancing topic sparsity and coherence (Srivastava & Sutton 2017). The model learns topic vectors that represent distinct aspects of the data, which are combined multiplicatively to form the overall topic distribution, resulting in more focused and interpretable topics compared to standard VAEs.

While NTMs like NVDM and ProdLDA offer greater flexibility by employing VAEs to learn nonlinear mappings and capture intricate document-topic structures, they also have notable drawbacks. They introduce substantial computational complexity, requiring extensive training and hyperparameter tuning, which can be resource intensive. Furthermore, their reliance on deep neural networks makes them less interpretable than traditional topic models, and they can be challenging to scale for very large corpora. Despite these challenges, NTMs provide a more expressive framework for topic modeling, capable of uncovering nuanced patterns in text data.

Despite the advancements made by NTMs in capturing intricate patterns in text data, challenges persist in handling dominant words that frequently appear across topics, often obscuring less frequent but domain-specific terms. Techniques like term frequency - inverse document frequency (TF-IDF) help highlight rare words by weighing terms based on their inverse document frequency (Ramos et al. 2003), but don't influence the topic modeling process to ensure distinct and exclusive topics. This limitation is crucial in Dark Web discussion forums, where critical indicators often lie in rare vocabulary.

The Exclusive Topic Model (ETM) (Lei & Chen 2021) addresses this challenge through a weighted Lasso penalty and pairwise Kullback-Leibler (KL) divergence penalty. The weighted Lasso penalty penalizes frequently occurring words during topic estimation, reducing their impact on topic distributions. In ETM, words with higher frequencies are assigned greater penalties, which suppresses their influence and allows less frequent, field-specific terms to emerge more prominently. This contrasts with LDA, which lacks mechanisms to distinguish between frequent and rare terms, often resulting in topics dominated by common words.

To ensure that topics are distinct, ETM incorporates a pairwise KL divergence penalty into its optimization objective. This penalty enforces topic exclusivity by minimizing the overlap between topic distributions, leading to well-separated topics. The variational Expectation-Maximization (EM) algorithm used in ETM maximizes a penalized Evidence Lower Bound (ELBO), adjusting both local and global parameters to enhance topic coherence and separation. Simulation studies in the ETM paper show that it effectively mitigates the predominance of frequently appearing words and can also identify important but infrequently appearing words in the corpus. However, ETM increases computational complexity due to the weighted Lasso and KL divergence penalties, making optimization resource intensive. It also requires careful fine-tuning of parameters, which can be challenging and limits scalability for large datasets (Lei & Chen 2021).

As models like Transformers (Vaswani 2017) entered the literature, topic modeling began to harness the power of contextualized embeddings, enabling a deeper understanding of semantic relationships in text (Devlin et al. 2019). This shift led to the development of advanced topic modeling approaches that leverage pre-trained language models to generate more coherent and contextually rich topics (Bianchi et al. 2021).

BERTopic (Grootendorst 2022) is one such model that utilizes the power of Transformers, particularly BERT (Bidirectional Encoder Representations from Transformers), to generate document embeddings. Unlike traditional topic models, BERTopic leverages these contextual-

ized embeddings to capture the semantic nuances of text. It then applies a clustering algorithm, typically hierarchical density-based spatial clustering of applications with noise (HDBSCAN), to group similar documents into clusters representing topics. To extract the most representative words for each topic, BERTopic employs techniques like c-TF-IDF (class-based Term Frequency-Inverse Document Frequency), which identifies the most relevant words within each cluster by weighing terms based on their importance in the specific context of the topic. This combination of contextual embeddings and robust clustering allows BERTopic to produce highly coherent topics that are sensitive to the subtle semantic differences in text.

BERTopic's use of pre-trained models like BERT enables it to overcome many of the limitations of traditional topic models, such as ignoring word order and context. However, it has drawbacks. The reliance on pre-trained language models like BERT makes it computationally intensive, especially for large datasets, requiring substantial processing power and memory (Raffel et al. 2020). Additionally, BERTopic assumes that each document belongs to a single topic, which may not reflect the true thematic diversity of complex documents that span multiple topics. Another issue is the redundancy in topic distributions; because BERTopic relies on clustering and context-based term weighting, it can sometimes include redundant or semantically similar words within the same topic, reducing the clarity and distinctiveness of the topic's representation (Grootendorst 2022).

As topic modeling evolved with the integration of neural embeddings and contextualized language models, recent theoretical approaches aimed to further refine the extraction of coherent and diverse and address the drawbacks of previous embedding-based topic model.

S3 (Kardos et al. 2024) reframes topic modeling as a signal separation problem using Independent Component Analysis (ICA) (Hyvärinen & Oja 2000). It directly utilizes contextual embeddings, such as those generated by sentence transformers (Reimers 2019), to identify latent semantic components that represent document-topic importance. By decomposing document embeddings into a mixing matrix and a source matrix through ICA, S3 identifies distinct semantic axes, each representing a topic. This process results in a term-topic matrix where the importance of terms is estimated based on their relationships to these semantic axes, offering a coherent and direct representation of topics within the embedding space. Unlike models like BERTopic, which assume a single topic per document, S3 can associate multiple topics with a single document, providing a more flexible framework that mirrors the complex structure of real-world texts.

A distinctive feature of S3 is its ability to accommodate negative term importances, adding a "negative definition" for topics by highlighting not only what a topic includes but also what it excludes. While S3 is a relatively new model offering a fresh perspective, it has some challenges. It assumes symmetrical importance between terms and components, potentially leading to redundancy in topic descriptions. The model's performance heavily relies on the quality of underlying sentence embeddings, making it sensitive to the semantic nuances of the pre-trained model. Additionally, the use of negative term importances, although enriching, can complicate topic interpretation.

The above literature review does not encompass all topic models within the research domain. However, it focuses on the most comprehensive and foundational models relevant to the research and topic modeling as a whole. To highlight the key topic models discussed, Table 1 provides a concise overview of their main contributions and limitations.

Model	Year	Key Contribution	Overlapping Limitations
NMF	1999	Linear algebraic factorization	• Lacks probabilistic foundation • Predefined number of topics required. • Limited interpretability.
pLSI	1999	Probabilistic approach, word mixtures	• Overfitting • Poor generalization to new documents • Limited scalability
LDA	2003	Probabilistic generative model	• Requires predefined number of topics • Bag-of-words assumption • Ignores word order • Struggles with rare words • Topic coherence issues
HDP	2006	Nonparametric Bayesian model, infers number of topics from the data.	• Computationally intensive • Relies on bag-of-words • Ignores context and word order
CTM	2006	Models topic correlations	• Ignores word order and context • Computationally complex • Less interpretable than simpler models
DTM	2006	Temporal evolution of topics	• Inherits LDA's limitations • Complex to implement • Difficult to capture short-term fluctuations
Online LDA	2010	Scalable LDA with stochastic inference	• Relies on bag-of-words assumption • Parameter sensitivity • Limited to predefined topics
BTM	2013	Topic modeling for short texts	• Misses broader document context • Limited to short texts • Struggles with nuanced semantics
LDA2Vec	2016	Combines LDA with static word embeddings	• Fails to capture word sense variations since it uses static pretrained word embeddings • Computationally demanding
NVDM	2016	Uses VAEs for continuous latent space	• High computational complexity • Hard to interpret • Requires extensive training
ProdLDA	2017	Enhances topic sparsity	• Resource-intensive training • Less interpretable than traditional models • Requires significant hyperparameter tuning
ETM	2021	Weighted Lasso and KL divergence for topic exclusivity	• Computational complexity • Requires careful fine-tuning • Challenging for large datasets
BERTopic	2022	Contextual embeddings with BERT	• Computationally intensive • High memory usage • Redundancy in topic terms • Assumes single topic per document
S3	2024	ICA for signal(Topic) separation in embeddings	• Redundancy in topic descriptions • Complex interpretation • Sensitive to the quality of embeddings • Recently developed, not yet thoroughly reviewed in the literature.

Table 1: Comparative Overview of Topic Modeling Approaches with Key Contributions and Overlapping Limitations

3 Project Management

3.1 Project Plan

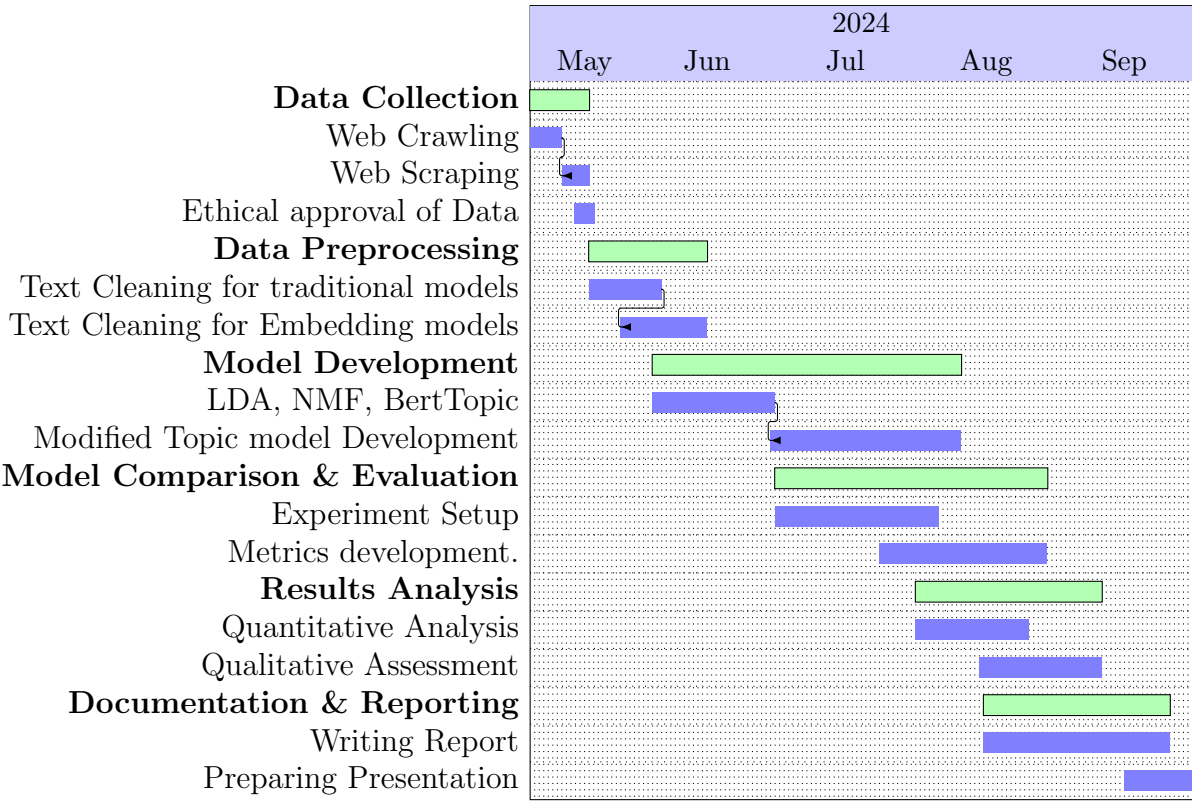


Figure 1: Gantt Chart

The project followed a structured management plan, with key phases outlined and tracked through a Gantt chart refer figure 1.(Figure 1 is inspired by and largely derived from my previously submitted research proposal for this project) The timeline was developed to ensure timely completion, accounting for potential challenges in data collection, preprocessing, model development, evaluation, and reporting.

3.1.1 Project Process

The project was divided into key phases, each contributing to the development of the final product, a comparative topic modeling analysis for dark web discussion forums, resulting in insights into the most effective models for detecting low-frequency but important terms. The original process followed these steps:

- Data collection - This phase involved finding clear web mirrors of Dark web forums. And thoroughly understand the architecture and HTML structure for further data scraping. and after data collection ethical approval of the data scraped.
- Data Preprocessing - This phase was split between traditional models and Embedding based model. To prepare the data for further development.
- Model Development - This phase was allocated for model development of LDA, NMF and BERTopic followed by finding research gap and modifying existing model.

- Model Comparison and Evaluation - This phase included setting up the experimental structure and environment, and metric development for comparison.
- Results & evaluation - This phase was associated with quantitative analysis and qualitative assessment of topic models.

3.1.2 Managing Risks, Challenges, and Adjustments

At the start of the project, several risks were identified that needed careful attention and adjustments as the work progressed. The first challenge was Data Collection, where finding reliable web mirrors and archives for Dark Web forums turned out to be more time-consuming than expected due to the limited availability of such sources. To handle this, extra time was included in the plan to deal with any technical issues or increased computational needs. The data scraping process was also quite complex, as it required understanding the architecture and HTML structure of the forums.

In the Data Preprocessing phase, there was a concern about accidentally filtering out important, less frequent terms due to the aggressive noise-cleaning methods. To manage this, sample data was manually reviewed, and specific regular expressions were developed to remove noise. However, these patterns couldn't recognize and handle "*leetspeak*" (like "D4RK W3B"), which posed a risk of losing key terms in the process.

During Model Development, there were challenges related to the heavy computational resources needed, especially for embedding-based models that require a lot of processing power. To mitigate this, the training process was optimized by using smaller batch sizes and simplifying the models where possible. Extra time was also built into the schedule in case these computational needs led to delays.

Although the project mostly followed the initial timeline, a few adjustments were necessary. Data Collection took slightly longer due to the difficulties in finding appropriate sources and the complexity of scraping. This delay was handled using the extra time planned for such challenges. In the Model Development phase, unforeseen computational limitations meant that the buffer time had to be used to keep things on track.

Regular meetings with my supervisor helped me in managing the project. Their feedback helped refine the research objectives and adjust the approach where needed. These discussions provided valuable insights, ensuring that the project stayed aligned with its goals while allowing flexibility to address unexpected challenges.

4 Topic Modelling Techniques

There are many topic modeling approaches, but because the domain and data in this research are unique, it's important to explore multiple options. Choosing the right technique isn't always straightforward in NLP, it often involves a lot of trial and error, as researchers test various models to find the one that works best for the specific data.

The initial phase of this research, which involved applying LDA, has been published in the IEEE ISSE paper titled '*DMiNority Report: On Monitoring Cyber-crime Innovation in Dark Net Markets*,' co-authored by Dr. Vitor Jesus, myself, and colleagues (Jesus et al. 2024). Under Dr. Jesus's supervision, I contributed by applying LDA to analyze topics within the Dread forum, which laid the groundwork for this dissertation. Building on that foundation, this dissertation dives deeper into exploring topic modeling techniques, particularly looking at how models like LDA, NMF, and semantic rich embedding approach like BERTopic work, and how they can be adapted or modified to better capture low-frequency, high-impact terms and evolving trends within the Dark Web.

4.1 LDA

1. Problem Setup

Given a corpus of M documents and a vocabulary of V unique words, LDA tries to uncover K latent topics. The goal is to represent each document as a mixture of topics and each topic as a distribution over words. The key components of the model are:

- **Document-Topic Distribution** (θ_d): Each document d is associated with a probability distribution over topics.
- **Topic-Word Distribution** (ϕ_k): Each topic k is associated with a probability distribution over words.
- **Topic Assignment for Words** ($z_{d,n}$): Each word in a document is assigned to a topic.
- **Word in Document** ($w_{d,n}$): The n -th word in document d .

2. Generative Process of LDA

Document Generation Process:

- (a) For each topic k , LDA assumes the words are drawn from a multinomial distribution ϕ_k , where the parameters ϕ_k are drawn from a Dirichlet distribution with parameter β :

$$\phi_k \sim \text{Dirichlet}(\beta) \quad (1)$$

- (b) For each document d , LDA assumes the topics are drawn from another multinomial distribution θ_d , with parameters θ_d drawn from a Dirichlet distribution with parameter α :

$$\theta_d \sim \text{Dirichlet}(\alpha) \quad (2)$$

- (c) For each word $w_{d,n}$ in document d :

- A topic $z_{d,n}$ is chosen from θ_d , the document's topic distribution:

$$z_{d,n} \sim \text{Multinomial}(\theta_d) \quad (3)$$

- A word $w_{d,n}$ is drawn from $\phi_{z_{d,n}}$, the word distribution of the chosen topic:

$$w_{d,n} \sim \text{Multinomial}(\phi_{z_{d,n}}) \quad (4)$$

3. Inference and Posterior Estimation

The primary goal in LDA is to infer the latent variables—the topic assignments z , the topic-word distributions ϕ , and the document-topic distributions θ —given the observed words w in the documents. However, calculating the exact posterior distribution $p(z, \theta, \phi | w, \alpha, \beta)$ is intractable because it requires summing over all possible topic assignments and integrating over continuous distributions.

4. Approximate Inference: Gibbs Sampling

Since exact inference is difficult, LDA uses approximate methods like Gibbs Sampling, a Markov Chain Monte Carlo (MCMC) technique, to estimate the posterior distribution. Gibbs Sampling iteratively updates the topic assignment for each word, conditioned on the current topic assignments of all other words. The key equation for sampling the topic $z_{d,n}$ is:

$$p(z_{d,n} = k | z_{-d,n}, w, \alpha, \beta) \propto (n_{d,k}^{-d,n} + \alpha) \cdot \frac{n_{k,w_{d,n}}^{-d,n} + \beta}{n_k^{-d,n} + V\beta} \quad (5)$$

where:

- $n_{d,k}^{-d,n}$ is the number of words in document d assigned to topic k , excluding the current word.
- $n_{k,w_{d,n}}^{-d,n}$ is the number of times word $w_{d,n}$ has been assigned to topic k , excluding the current word.
- $n_k^{-d,n}$ is the total number of words assigned to topic k , excluding the current word.

5. Extracting Distributions

After running Gibbs Sampling for several iterations, we can compute:

- **Topic-Word Distribution** (ϕ_k):

$$\phi_{k,v} = \frac{n_{k,v} + \beta}{n_k + V\beta} \quad (6)$$

This tells us how likely word v is to appear in topic k .

- **Document-Topic Distribution** (θ_d):

$$\theta_{d,k} = \frac{n_{d,k} + \alpha}{N_d + K\alpha} \quad (7)$$

This tells us the proportion of topic k in document d .

4.2 NMF

1. Problem Setup

Given a corpus of M documents and a vocabulary of V unique words, NMF represents the document-term matrix $\mathbf{X} \in \mathbb{R}^{M \times V}$, where X_{ij} indicates the frequency of word j in document i . The goal is to factor $\mathbf{X}$ into:

- **Document-Topic Matrix** $\mathbf{W} \in \mathbb{R}^{M \times K}$: Each row represents a document's contribution to the K topics.
- **Topic-Word Matrix** $\mathbf{H} \in \mathbb{R}^{K \times V}$: Each row represents the word distribution over the K topics.

The factorization seeks to approximate $\mathbf{X}$ as:

$$\mathbf{X} \approx \mathbf{WH} \quad (8)$$

with the constraint that all elements in $\mathbf{W}$ and $\mathbf{H}$ are non-negative ($W_{ik} \geq 0$ and $H_{kj} \geq 0$).

2. Matrix Factorization Model

NMF aims to minimize the reconstruction error between the original document-term matrix $\mathbf{X}$ and its approximation $\mathbf{WH}$. The objective function is:

$$\min_{\mathbf{W}, \mathbf{H}} \|\mathbf{X} - \mathbf{WH}\|_F^2 \quad (9)$$

where $\|\cdot\|_F$ is the Frobenius norm, which sums the squares of the differences between the original matrix and its approximation. This measures how well the factorization approximates the original data.

3. Optimization with Multiplicative Update Rules

NMF employs an iterative optimization algorithm to update the values of $\mathbf{W}$ and $\mathbf{H}$. One of the most common methods is the multiplicative update rule, which ensures that both matrices remain non-negative during the updates.

- **Update for $\mathbf{W}$:**

$$W_{ik} \leftarrow W_{ik} \frac{(\mathbf{X}\mathbf{H}^T)_{ik}}{(\mathbf{W}\mathbf{H}\mathbf{H}^T)_{ik}} \quad (10)$$

- **Update for $\mathbf{H}$:**

$$H_{kj} \leftarrow H_{kj} \frac{(\mathbf{W}^T\mathbf{X})_{kj}}{(\mathbf{W}^T\mathbf{W}\mathbf{H})_{kj}} \quad (11)$$

These rules alternate between updating $\mathbf{W}$ and $\mathbf{H}$, ensuring non-negativity and gradually minimizing the reconstruction error.

4. Extracting Topics

Once the matrices $\mathbf{W}$ and $\mathbf{H}$ converge, they provide insights into the topics and their distributions:

- **Document-Topic Matrix $\mathbf{W}$:**
 - Each row W_i shows the contribution of different topics to document i .
 - A high value of W_{ik} means topic k is strongly represented in document i .
- **Topic-Word Matrix $\mathbf{H}$:**
 - Each row H_k shows the contribution of different words to topic k .
 - A high value of H_{kj} indicates word j is important in topic k .

4.3 Bertopic

1. Problem Setup

Given a collection of M documents, BERTopic models the topics by first generating dense document embeddings using transformer-based models, such as BERT. The main steps involve:

- **Generating embeddings** that encode the semantic meaning of documents.
- **Clustering embeddings** to group similar documents into clusters.
- **Extracting representative words** from each cluster to define the topics.

This modular approach allows users to customize each stage, from embedding generation to topic representation, using a variety of algorithms and techniques.

2. Model Description and Key Components

BERTopic consists of several key stages that are tightly integrated:

- **Document Embeddings:** Each document d_i is transformed into a high-dimensional vector, $\mathbf{e}_i$, using pre-trained transformer models like BERT:

$$\mathbf{e}_i = \text{BERT}(d_i) \quad (12)$$

These embeddings capture the contextual and semantic richness of the text, forming the basis for clustering.

- **Dimensionality Reduction:** Since BERT embeddings are high-dimensional, dimensionality reduction techniques such as UMAP (McInnes & Healy 2018) are applied to project the embeddings into a lower-dimensional space:

$$\mathbf{E}' = \text{UMAP}(\mathbf{E}) \quad (13)$$

UMAP lowers noise and preserves the global data structure, thus enabling effective clusterings.

- **Clustering:** The reduced embeddings $\mathbf{E}'$ are grouped into clusters utilizing HDBSCAN, a density-based clustering algorithm capable of identifying clusters with different densities without requiring the predefinition of the number of clusters (Campello et al. 2013):

$$C = \text{HDBSCAN}(\mathbf{E}') \quad (14)$$

HDBSCAN automatically detects the most meaningful clusters and assigns documents accordingly, even allowing some documents to be labeled as noise if they do not fit well into any cluster.

- **Topic Representation:** Once clusters $C = \{c_1, c_2, \dots, c_K\}$ are identified, BERTopic uses class-based TF-IDF (c-TF-IDF) (Grootendorst 2022) to extract representative words for each cluster:

$$\text{c-TF-IDF}(w_j, c_k) = \text{TF}(w_j, c_k) \times \log \left(\frac{N}{|C(w_j)|} \right) \quad (15)$$

Here, $\text{TF}(w_j, c_k)$ is the term frequency of word w_j in cluster c_k , N is the total number of clusters, and $|C(w_j)|$ is the number of clusters containing the word w_j . This ensures that words are not only representative of their cluster but also discriminative compared to other clusters.

3. Extracting Topics

Once clustering and topic representation are complete, topics are extracted as follows:

- **Identifying Top Words:** For each cluster c_k , BERTopic identifies the words with the highest c-TF-IDF scores, which are used to label and interpret the topic.
- **Topic-Document Association:** Each document is assigned to a topic based on its cluster membership. The distance of a document from the cluster center in the embedding space indicates the document's strength of association with the topic.

4.4 Contribution to the research

LDA treats all words equally in its prior distributions, which can under represent the rare but informative words especially in specialized domains where such terms are crucial.

To address this limitation, we introduce a **Modified LDA model with Dynamic Word Influence**. This adaptation enhances the standard LDA by adjusting the influence of words based on their rarity, giving prominence to rare and exclusive terms during topic generation. Importantly, while we modify the way word influence is incorporated, the core generative process and assumptions of LDA remain unchanged.

To the best of our knowledge, incorporating dynamic word influence directly into the Dirichlet prior for the topic-word distributions is a novel approach that offers a new perspective on emphasizing rare words in topic modeling.

1. Modified LDA Model Overview

Our Modified LDA retains the core structure and generative assumptions of the standard probabilistic model where documents are mixtures of topics, and topics are mixtures of words. The key innovation is the introduction of **dynamic word-specific priors** in the topic-word distribution. We compute a **Word Influence (WI)** score for each word, adjusting the Dirichlet prior β accordingly.

By keeping the fundamental mechanics of LDA unchanged, we ensure that our model remains compatible with existing inference techniques and theoretical foundations.

2. Computing Word Influence

(a) Formula

The **Word Influence (WI)** for each word w is calculated as:

$$WI_w = \lambda \times \ln \left(\frac{N + \epsilon}{1 + df_w} \right) \quad (16)$$

- λ : A scaling factor controlling the emphasis on rare words.
- N : Total number of documents in the corpus.
- ϵ : A small constant to prevent division by zero.
- df_w : Document frequency of word w , i.e., the number of documents containing word w

This formula is inspired by the Inverse Document Frequency (IDF) component of TF-IDF weighting, where words that appear in fewer documents receive higher scores. The use of the logarithmic function and the ratio of total documents to document frequency mirrors the standard IDF calculation, as described by [Salton & Buckley \(1988\)](#)

(b) In the code, this computation is implemented as follows:

```

1 def compute_document_frequency(corpus):
2     df_counts = defaultdict(int)
3     for document in corpus:
4         unique_words = set(document)
5         for word in unique_words:
6             df_counts[word] += 1
7     return df_counts
8
9 def compute_word_influence(df_counts, D, lambda_factor=2, epsilon=1e
10    -5):
11     """
12     lambda_factor can be adjusted to control the degree of emphasis on
13     rare words.
14     """
15     word_influence = {
16         word: lambda_factor * np.log((D + epsilon) / (1 + count))
17         for word, count in df_counts.items()
18     }
19     return word_influence

```

Listing 1: Document Frequency and Word Influence Calculation

- `compute_document_frequency` : Calculates df_w by counting how many documents each word appears in.
- `compute_word_influence` : Applies the mathematical formula to compute WI_w for each word using the total number of documents N .

3. Incorporating Word Influence into the Model

(a) Modified Dirichlet Prior

We adjust the topic-word Dirichlet prior β to be word-specific:

$$\beta_w = \beta \times WI_w \quad (17)$$

- β_w : The adjusted prior for word w .

- β : The base prior scalar.

By modifying β in this way, we emphasize rare words within the Bayesian framework, maintaining the properties of the Dirichlet distribution.

(b) **Implementation to compute β_w in the code:**

```

1  "
2  V: Vocabulary size, i.e., the number of unique words.
3  "
4
5  def get_word_influence_array(word_influence, V):
6      word_influence_array = np.zeros(V)
7      for word_id, influence in word_influence.items():
8          word_influence_array[word_id] = influence
9      return word_influence_array
10
11 def compute_beta_w(beta, word_influence_array):
12     beta_w = beta * word_influence_array
13     beta_sum = np.sum(beta_w)
14     return beta_w, beta_sum

```

Listing 2: Computing β_w

- `get_word_influence_array` : Converts the word influence dictionary into an array aligned with word indices (from 0 to $V - 1$)
- `compute_beta_w` : Calculates β_w for each word and computes the total sum $\beta_{\text{sum}} = \sum_{w=1}^V \beta_w$ for normalization.

(c) **Generative Process**

The rest of the LDA model, including the document-topic prior α and the generative process, remains unchanged, with the only difference being the word-specific priors in the topic-word distributions.

Variables:

- K : Number of topics.
- ϕ_k : Topic-word distribution for topic k .
- θ_d : Document-topic distribution for document d .
- N_d : Number of words in document d .
- $z_{d,n}$: Topic assignment for the n^{th} word in the document d .
- $w_{d,n}$: Word at position n in the document d .
- α : Hyperparameter for the Dirichlet prior on θ_d .

i. **For each topic k ($k = 1, 2, \dots, K$):**

- Draw ϕ_k from a Dirichlet distribution with parameters β_w :

$$\phi_k \sim \text{Dirichlet}(\beta_1, \beta_2, \dots, \beta_V)$$

Each β_w is specific to word w , reflecting its influence.

ii. **For each document d ($d = 1, 2, \dots, N$):**

- A. Draw θ_d from a Dirichlet distribution:

$$\theta_d \sim \text{Dirichlet}(\alpha)$$

- B. For each word position n ($n = 1, 2, \dots, N_d$):

- Sample topic $z_{d,n}$ from:

$$z_{d,n} \sim \text{Multinomial}(\theta_d)$$

- Sample word $w_{d,n}$ from:

$$w_{d,n} \sim \text{Multinomial}(\phi_{z_{d,n}})$$

The modifications introduced in the model primarily adjust the way word distributions are handled by dynamically changing the Dirichlet prior β to β_w , giving more weight to rare words. However, the core structure of LDA remains unchanged. The document-topic prior α , the process of sampling θ_d , and the overall generative nature of the model are preserved.

4. Justification

(a) Bayesian Framework Consistency

- **Dirichlet-Multinomial Conjugacy:** The Dirichlet distribution is conjugate to the Multinomial distribution. Adjusting β to β_w maintains this conjugacy, allowing for tractable posterior computations. This is essential for efficient inference, as highlighted in [Heinrich \(2005\)](#).
- **Preservation of Core Assumptions:** By only modifying the priors while keeping the likelihood and the rest of the model intact, we ensure that the fundamental probabilistic structure of LDA is preserved.

(b) Probabilistic Validity

- **Valid Priors:** Each $\beta_w > 0$, ensuring the Dirichlet prior remains valid.
- **Normalization:** The Dirichlet and Multinomial distributions normalize over their support, maintaining proper probability distributions.
- **Consistency in Inference:** The standard inference techniques used in LDA, such as Collapsed Gibbs Sampling, remain applicable.

5. Inference via Collapsed Gibbs Sampling

To infer the latent variables, we use **Collapsed Gibbs Sampling**, integrating out θ and ϕ to focus on sampling the topic assignments z . This method remains the same as in standard LDA, with adjustments to account for the word-specific priors.

(a) Sampling Equation

The probability of assigning word w in document d to topic k is:

$$p(z_{d,n} = k | z_{-d,n}, w, \alpha, \beta_w) \propto (n_{d,k}^{-d,n} + \alpha) \times \frac{n_{k,w}^{-d,n} + \beta_w}{n_k^{-d,n} + \sum_v \beta_v} \quad (18)$$

- $n_{d,k}^{-d,n}$: Count of words in document d assigned to topic k , excluding current word.
- $n_{k,w}^{-d,n}$: Count of word w assigned to topic k , excluding current word.
- $n_k^{-d,n}$: Total words assigned to topic k , excluding current word.

This equation is derived following the methodology in [Griffiths \(2004\)](#), adapted to include the word-specific prior β_w .

(b) Interpretation

- **First Term** ($n_{d,k}^{-d,n} + \alpha$): Preference for topics prevalent in the document.
 - **Second Term** ($n_{k,w}^{-d,n} + \beta_w$): Preference for assigning word w to topics where it's prominent, adjusted by its influence.
- (c) The Gibbs Sampling algorithm is adjusted to incorporate the word-specific priors in code :

```

1 def gibbs_sampling(corpus, K, V, alpha, beta, iterations, beta_w,
2   beta_sum, n_d_k, n_k_w, n_k, topic_assignments):
3     """
4     Gibbs sampling for LDA with dynamic word influence.
5
6     Variables:
7     n_d_k: Document-topic count matrix (D
8     * K).
9     n_k_w: Topic-word count matrix (K * V).
10    n_k: Topic count vector (K).
11    topic_assignments: List of topic assignments for each word in
12    each document.
13    iterations: Number of Gibbs sampling iterations.
14    """
15    D = len(corpus)
16    for it in range(iterations):
17        for d, doc in enumerate(corpus):
18            for i, word in enumerate(doc):
19                current_topic = topic_assignments[d][i]
20                n_d_k[d, current_topic] -= 1
21                n_k_w[current_topic, word] -= 1
22                n_k[current_topic] -= 1
23
24                denominator = n_k + beta_sum + 1e-10
25                log_topic_probs = (
26                    np.log(n_d_k[d, :] + alpha) +
27                    np.log(n_k_w[:, word] + beta_w[word]) -
28                    np.log(denominator)
29                )
30
31                # logsumexp for numerically stable log probability
32                # calculations
33                topic_probs = np.exp(log_topic_probs - logsumexp(
34                    log_topic_probs))
35
36                new_topic = np.random.choice(K, p=topic_probs)
37                topic_assignments[d][i] = new_topic
38                n_d_k[d, new_topic] += 1
39                n_k_w[new_topic, word] += 1
40                n_k[new_topic] += 1
41    return n_d_k, n_k_w, n_k, topic_assignments

```

Listing 3: Adjusted Gibbs Sampling Algorithm

The sampling equation incorporates β_w and β_{sum} , accounting for the word-specific priors. The updated probabilities reflect the influence of rare words through β_w . The rest of the Gibbs Sampling process follows the standard approach as described in Griffiths (2004).

6. Estimating Distributions

After convergence, we estimate:

- **Topic-Word Distribution ϕ_k :**

$$\phi_{k,w} = \frac{n_{k,w} + \beta_w}{n_k + \sum_v \beta_v} \quad (19)$$

Represents the probability of word w given topic k , adjusted by the word's influence.

- **Document-Topic Distribution θ_d :**

$$\theta_{d,k} = \frac{n_{d,k} + \alpha}{N_d + K\alpha} \quad (20)$$

Where N_d is the the total number of words in document d .

Represents the probability of topic k in document d .

- **These functions in code can be written as :**

```

1 def compute_phi(n_k_w, beta_w, n_k, beta_sum):
2     """Compute topic-word distribution phi."""
3     return (n_k_w + beta_w[np.newaxis, :]) / (n_k[:, np.newaxis] +
4         beta_sum)
5
6 def compute_theta(n_d_k, alpha, K):
7     """Compute document-topic distribution theta."""
8     return (n_d_k + alpha) / (np.sum(n_d_k, axis=1)[:, np.newaxis] +
9         K * alpha)

```

Listing 4: Estimation of Distributions.

- `compute_phi`: Calculates the topic-word distributions ($K \times V$ matrix).
- `compute_theta`: Calculates the document-topic distributions ($N \times K$ matrix).

These estimations are consistent with the original LDA model, with the only difference being the word-specific prior β_w .

The adjustments fit within the Bayesian framework, maintaining Dirichlet-multinomial conjugacy and preserving the model's probabilistic structure. The inference process via Collapsed Gibbs Sampling remains consistent with standard LDA, ensuring that the model is both theoretically sound and practically implementable.

For a complete implementation of the Modified LDA model, please refer to the [GitHub Repository](#)¹.

5 Methodology

5.1 Data Source, Data collection, Data overview.

Several efforts have been made to collect Dark Web data through direct scraping and archiving to create static datasets. An example of this is the CrimeBB dataset provided by the Cambridge Cybercrime Centre, which collected data from four different active forums using a crawler called CrimeBot (Pastrana et al. 2018). Access to the CrimeBB dataset, however, requires signing a legal agreement due to the sensitive nature of the content. Additionally, Branwen et al. (2015) provided darknet market archives, but there is limited information regarding the ethical considerations involved in the data collection process. These initiatives highlight the challenges and complexities of obtaining and using Dark Web data. In this research, the data source and the methodology for collection and preprocessing will be elaborated in further sections.

¹<https://github.com/k4r71k-repo/Dissertation/tree/main>

5.1.1 Data Source

Considering connectivity using Tor can be slow and unreliable (often because of the servers themselves), we used CafeDread² as a clear web mirror of Dread forum on dark web. CafeDread.net provides read-only access to the latest posts and comments from Dread which is designed with a “Reddit-like” interface. This allowed us to get genuine dark web forum data but on clear web ethically. Refer figure 2 for the home page architecture. The site has a simple layout with a navigation bar at the top, including options like “New,” “Discover,” and “About.” Discover is the section with our scraping interest. The main section of the page displays posts. On the right side, there is a list of top communities, resembling a forum or discussion board layout, with various community names related to darknet topics. A search bar is located at the top right corner for browsing through the site’s content.

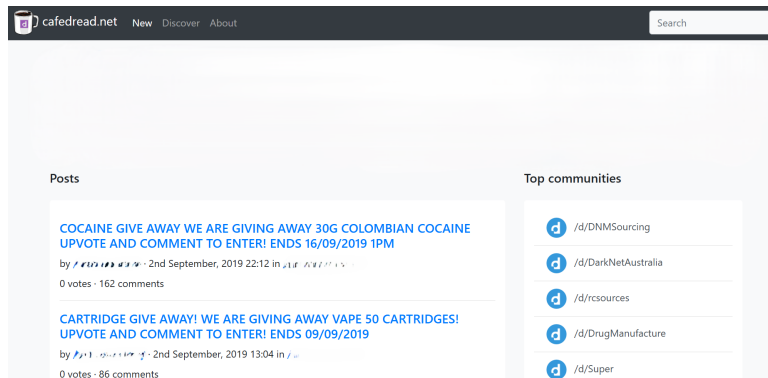


Figure 2: cafedread website

5.1.2 Data Collection

This clear web mirror facilitated our data collection process in a manner that aligns with ethical guidelines. By providing a public, accessible version of the Dread forum, CafeDread enabled us to gather data without directly interacting with the Dark Web or the individuals involved in those forums. By leveraging CafeDread, we maintained compliance with ethical standards by collecting publicly available data in a controlled and transparent manner, thereby minimizing the risks associated with direct Dark Web scraping.

The data scraping process was implemented using the [Scrapy](https://scrapy.org/)³ framework, designed to systematically extract structured information from the website “https://cafedread.net”. The primary goal was to capture data related to community discussions, including community titles, post details, authorship, timestamps, and content. The scraping mechanism is composed of a custom spider and a data pipeline for storing the extracted information. The HTML elements of the page were first thoroughly learnt for further process.

Scraper Architecture

1. Spider Design: This is the core component that performs the web scraping. It contains logic to navigate the website, extract relevant data, and handle pagination. The spider works in three main stages:
 - Stage 1: *Community Discovery*: The spider starts at the website’s discovery page (“Discover”). It uses `XPath` selectors to identify and collect links to different communities. For each community, it stores the community’s title and its link, then sends a request to parse the community page.

²cafedread.net

³<https://scrapy.org/>

- Stage 2:*Post Extraction*: On each community page, the spider finds links to individual posts using **XPath** selectors. It captures the post titles and URLs, then sends requests to follow these links to the detailed post pages.
- Stage 3:*Post Details Extraction*: The post's details container like "time", spider finds and extracts information, using **XPath** queries. This information is structured into an item for later processing.

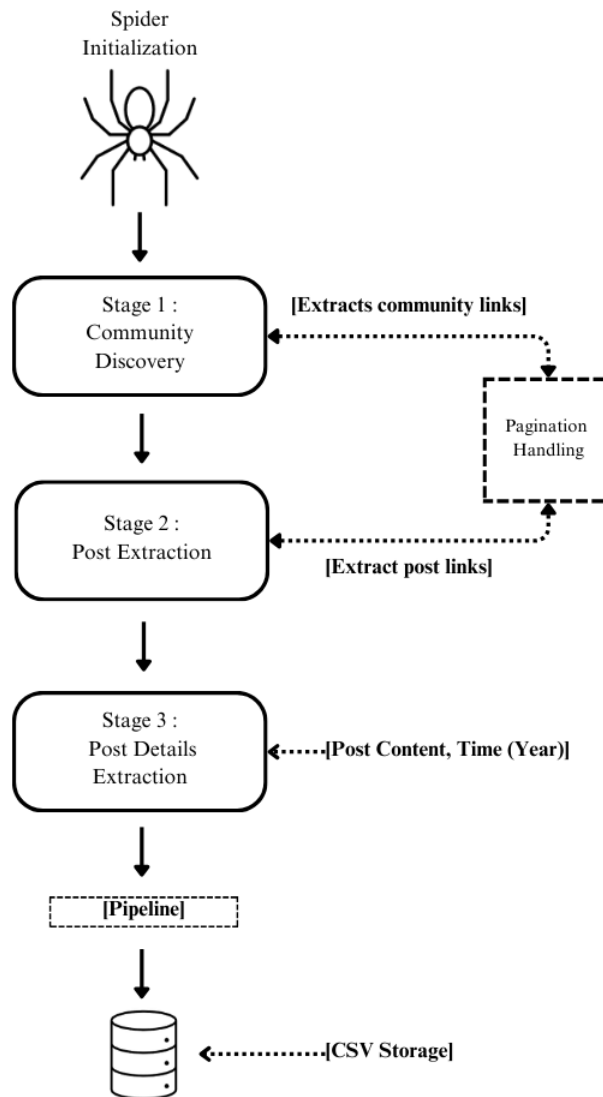


Figure 3: Scraper Architecture

2. Data Item Structure:

- Data Model (CommunityItem): This defines the structure of the data to be scraped. The fields are defined. Then These fields define what information will be extracted and stored for each post. The **TakeFirst** processor from scrapy is used to ensure that only the first matched result is taken when extracting data, preventing lists of values when only a single value is expected.

3. Data Flow and Requests:

- Request Flow:

- The spider sends HTTP requests to various pages of the website (communities, posts, etc.).
- It uses meta to pass information (e.g., `community_title` and `post_title`) between requests. This allows the spider to carry context from one page to the next, ensuring that details like which community a post belongs to are not lost as the spider navigates deeper into the site.
- **Pagination Handling:** Both in the community listing and the posts listing, the spider identifies "next page" links with `Xpath` and follows them, allowing it to systematically scrape all available content across multiple pages.

Data Extraction and Storage

1. Data Extraction:

- **Xpath Selectors:** The spider uses `Xpath` selectors to locate specific elements on the web pages. For instance, it identifies the community links, post titles, author names, publication dates, and post content by searching for specific HTML structures (e.g., classes and tags) that contain this information.
- **Item loader:** The spider uses an `ItemLoader` to populate the `CommunityItem` with extracted data. The item loader ensures that the extracted data is structured properly before it is passed on for storage.

2. Data Storage:

- **Pipeline:** This component manages how the scraped data is processed and saved. It opens a CSV file when the spider starts and writes data into this file as the spider extracts it.
- **Data Writing:** For each item (post) extracted by the spider, the pipeline writes a row in the CSV file containing the post's community title, post title, author, date, and content.
- **Finalization:** The pipeline ensures that the CSV file is properly closed once the spider completes its scraping process, preventing data loss or file corruption.

5.1.3 Data Overview

Initially, We collected a total of 22982 posts spanning 7 years. After collecting the data, preliminary manual data validation check was performed. After validation check we decided what to remove from the dataset and what to keep. We keep only `post_content` and `year` column for further data preprocessing. We initially cleaned the texts, we used regex to remove repeating patterns that did not contain any meaningful content. For example, PGP (Pretty Good Privacy) keys are used primarily for securing communications and ensuring data integrity. Then removed URLs, system messages and edit notes. Filtering out rows where the content column contained only placeholder text like "[removed]" or "[System Message] Invalid BBCODE Formatting.". And removed null values. Dread forum is mostly English but we decided to use langdetect and drop the posts which are not English from further analysis. After Preliminary data processing we had a total of 19661 posts for further analysis.

After performing the initial cleaning and filtering steps, we proceeded with further processing to prepare the data for detailed analysis and visualization. We tokenized each post using the SpaCy tokenizer, which helped in calculating both the total and unique token counts. The final token count across all posts in the dataset was 2,441,345, representing every individual token, including stopwords, punctuation, and special characters.

We also calculated the number of posts per year, which showed a sharp increase in activity starting in 2020. The number of posts per year grew significantly in 2023 and 2024, indicating a surge in engagement or possibly reflecting heightened discussions on emerging topics. Refer figure 4.

We then calculated the average token count per post per year to observe changes in post length. The analysis showed fluctuations, peaking in 2019, then decreasing before recovering in 2024. This variation could be due to fewer posts in early years giving consistent averages or shifts in discussion behavior, or site downtime, or inconsistent data collection. The exact cause of this fluctuation remains uncertain. Refer figure 5.

Next, we applied SpaCy's pre-trained English model for Named Entity Recognition (NER) to extract key entities, filtering out irrelevant content like PGP keys and ASCII symbols. The top entities, including "US," "UK," "EU," "MDMA," and "XMR" (Monero), highlight the platform's focus on geographic regions, cryptocurrencies, and drugs like MDMA (see figure 6). Notably, "Abacus" and "Archetyp" emerge as significant entities, pointing to major markets within the dark web. NER is crucial here, uncovering key topics, geopolitical trends, and market activities, providing valuable insights that could benefit businesses, cybersecurity firms monitoring dark web activity.

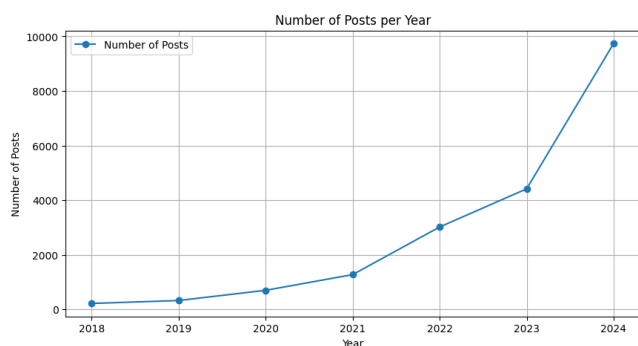


Figure 4: Number of Posts per Year

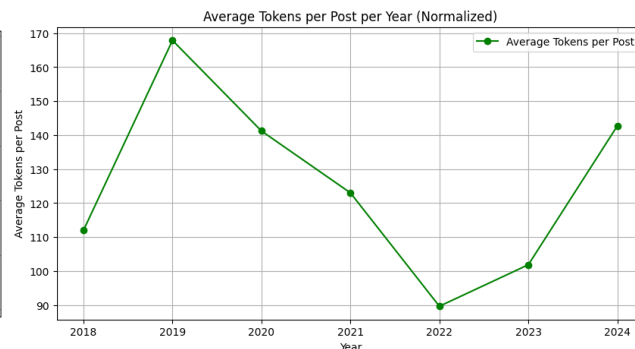


Figure 5: Average Tokens per Post per Year

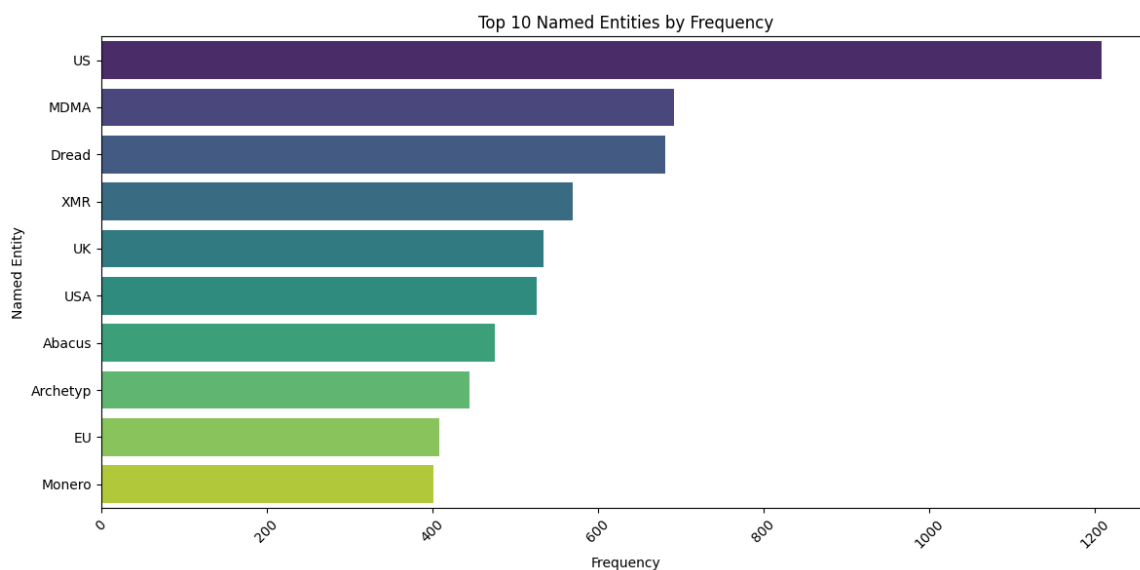


Figure 6: Named Entities Frequency count.

5.2 Experimental Setup for Topic Models

In this research, we compare five topic modeling techniques: baseline LDA, LDA with nouns and NER, Modified LDA, NMF with nouns and NER, and BERTopic. To ensure a fair comparison and draw accurate conclusions, it is crucial to standardize key experimental parameters across all models. This includes preprocessing steps, embedding generation (only BERTopic), hyperparameter used, and evaluation metrics. Each model has unique characteristics, but the core setup remains consistent to allow for meaningful insights into their comparative performance. Each model is applied yearly to extract yearly topics this is different from behaviour of dynamic topic modelling where the topics are tracked over the time. In contrast with our approach where we fit the model for every time span and analyse the dominant topics and model performance for that particular year.

Below, we describe the experimental setup for each of the topic models in detail, covering data preprocessing, model-specific configurations, and metrics used for evaluation.

5.2.1 Metrics

In this study, two key metrics are used to evaluate the performance of the topic models: Topic Diversity and Inverse Word Embedding Coherence (IWECC) Score. These metrics assess different aspects of the models, such as the uniqueness of words in the topics and the semantic coherence of those topics, respectively.

1. Topic Diversity (Dieng et al. 2020)

Topic Diversity measures the distinctiveness and variety of vocabulary among the principal terms derived from all topics produced by a topic model. The objective is to figure out whether the model generates diverse topics or reiterates identical terms across various topics. A greater diversity score signifies that topics are more varied and encompass a wider array of information.

The formula for topic diversity is:

$$\text{Topic Diversity} = \frac{\text{Number of Unique Words Across All Topics}}{\text{Total Number of Top Words Across All Topics}} \quad (21)$$

2. Internal Word Embedding Coherence Score.

Internal Coherence (Kardos et al. 2024) evaluates the effectiveness of a topic model in capturing the semantic relationships among words, tailored to the dataset in discussion. We evaluate this by calculating word-embedding-based coherence utilizing a FastText model trained on the identical corpus as the topic model. This technique, termed Inverse Word Embedding Coherence (IWECC), assesses the semantic similarity among the leading words in each topic by calculating the average pairwise cosine similarity of their embeddings. This method emphasizes the significant relationships among words instead of mere word co-occurrence, yielding a more precise assessment of topic coherence.

The formula for IWECC is:

$$\text{IWECC} = \frac{1}{\binom{n}{2}} \sum_{i=1}^{n-1} \sum_{j=i+1}^n \text{cosine_similarity}(w_i^t, w_j^t) \quad (22)$$

Where:

- n is the number of top words considered per topic.
- w_i^t and w_j^t are the word embeddings of the i -th and j -th top words in topic t .
- $\text{cosine_similarity}(w_i^t, w_j^t)$ is the cosine similarity between the embeddings of the two words.

5.2.2 Pre-processing

The preprocessing steps for the topic models in this research were designed to standardize the text input as much as possible across all models, ensuring a fair comparison. Common preprocessing steps include:

1. Removal of special characters and punctuation.
2. Elimination of single alphabet tokens, numbers, alphanumeric tokens, and PGP key patterns.
3. Lowercasing the text and removing stopwords using [NLTK's](https://www.nltk.org/)⁴ stopwords list.

These steps cleaned the dataset and ensured a consistent format for topic extraction. . However, for Noun NER LDA, Modified LDA and NMF, we extend preprocessing by introducing the extraction of nouns, proper nouns, and named entities (NER) using spacy. This allows these models to focus on key terms that are more likely to capture meaningful topics, particularly within domain-specific data.

In contrast, BERTopic takes a more simplified approach. As BERTopic leverages pre-trained transformer embeddings, it doesn't require noun or NER extraction. The preprocessing for this model is more lightweight, applying minimal text reduction since the context-aware embeddings capture semantic relationships.

By aligning preprocessing steps where possible and adjusting where necessary, we maintain a consistent comparison across models while leveraging the strengths of each approach.

5.2.3 LDA

1. Baseline LDA:

Corpus Preparation:

The preprocessed text data is tokenized into words, and a Bag-of-Words (BoW) representation is created for each document using Gensim's `Dictionary` and `doc2bow` method ([Řehůřek & Sojka 2010](#)). This BoW matrix serves as the input for the LDA model. The corpus consists of English posts extracted from the dataset, which are grouped by year to analyze how topics evolve over time.

Model Configuration:

The LDA model is trained on the corpus with the following parameters:

Parameter	Value
Number of Topics	10
Passes	10
Iterations	100
Random State	42
Top Words per Topic	20

Table 2: Parameter values for Baseline LDA.

The number of topics is fixed at 10, with 10 training passes and 100 iterations to ensure model convergence. A random state is set to 42 to make the results reproducible.

2. LDA with Nouns and NER:

⁴<https://www.nltk.org/>

For the LDA model with Nouns and NER, the preprocessing follows a similar procedure to the baseline LDA but slightly differs by focusing specifically on extracting nouns, proper nouns, and named entities (NER). This approach helps the model to concentrate on key terms likely to represent meaningful words in the topics.

Corpus Preparation: The dataset undergoes preprocessing similar to baseline, additionally nouns, proper nouns, and named entities (NER) are extracted using [spaCy⁵](#), resulting in a more focused corpus. After this extraction, the text is tokenized, and the Bag-of-Words (BoW) representation is generated. To further emphasize important and exclusive words in each document, the corpus is weighted using TF-IDF before being input into the LDA model.

Model Configuration:

Parameter	Value
Number of Topics	10
Passes	20
Iterations	200
Random State	42
Alpha α	'auto'
Eta η	'auto'
Top Words per Topic	10

Table 3: Parameter values for LDA with Nouns and NER.

The model is configured with an automatic optimization for the Dirichlet priors α (document - topic distribution) and η (topic-word distribution), allowing it to adapt to the corpus. The number of topics is fixed at 10, with 20 passes and 200 iterations to ensure proper convergence.

3. Modified LDA:

The Modified LDA model builds upon the baseline LDA by incorporating dynamic word influence into the topic-word Dirichlet prior β_w . This modification allows the model to adjust the weight of each word based on its frequency across the corpus, providing a mechanism to give more importance to rare, exclusive words, which are often underrepresented in traditional LDA.

The modified LDA model was created from scratch (see section 4.4) to include the custom word influence calculations.

In addition to that we apply this modified LDA model to each year's subset of documents to analyze topic evolution over time. We use a `year_column` in our dataset to filter documents and repeat the entire process for each year, ensuring that N (total number of documents in the corpus) represents the number of documents in each year's subset. Refer [GitHub Repository⁶](#) given in the footnote.

Corpus Preparation:

The text undergoes preprocessing, where nouns, proper nouns, and named entities (NER) are extracted using spaCy. After this, the corpus is tokenized, and a Bag-of-Words (BoW) representation is generated, following the same procedure as the baseline LDA. This

⁵<https://spacy.io/>

⁶<https://github.com/k4r71k-repo/Dissertation/tree/main>

preprocessing ensures that the input focuses on key entities, similar to the approach used in the LDA with Nouns and NER model.

Dynamic Word Influence Calculation:

The key modification in this model is the introduction of dynamic word influence. For each word in the corpus, its document frequency is calculated, and the influence of the word is computed based on a lambda factor λ , which amplifies the importance of rare words. This dynamic weighting is applied directly to the topic-word Dirichlet prior β_w , making the model focus more on unique terms that appear less frequently in the corpus.

Model Configuration:

The modified LDA model is configured with the following parameters:

Parameter	Value
Number of Topics	10
Passes	10
Iterations	100
Alpha	0.5
Beta (before modification)	0.1
Lambda Factor	10
Top Words per Topic	30

Table 4: Parameter values for Modified LDA.

The number of topics is set to 10, and the model performs 100 iterations of Gibbs sampling. The Lambda Factor is set to 10 to significantly scale up the effect of word influence. Other parameters are set to standard value. Beta is the standard Dirichlet prior for each word which is then influenced and controlled by the Lambda factor which scales the importance of rare words in the corpus. This explains Beta (before modification) in Table 4.

5.2.4 NMF

In the NMF model, the goal is to divide the term-document matrix into two non-negative matrices. One matrix will show how topics are distributed across words, and the other will show how documents are distributed across topics. The NMF approach allows for non-probabilistic, linear combinations of topics, making it distinct from LDA-based models. In this research we use `sklearn.decomposition.NMF`⁷ to build the NMF topic model.

Corpus Preparation:

The dataset undergoes preprocessing as described previously, focusing on cleaning and extracting nouns, proper nouns, and named entities using spaCy. The preprocessed text is tokenized, and a TF-IDF representation is created for each document. This TF-IDF matrix serves as the input to the NMF model, emphasizing the importance of exclusive and rare terms in the corpus, similar to other topic models used in this research.

Model Configuration:

The NMF model is configured with the following parameters:

⁷<https://scikit-learn.org/stable/modules/generated/sklearn.decomposition.NMF.html>

Parameter	Value
Number of Topics	10
Max Iterations	500
Top Words per Topic	30
Random State	42

Table 5: Parameter values for NMF

NMF is applied to the TF-IDF matrix, where the number of topics is set to 10. The maximum number of iterations for the algorithm is 500 to ensure convergence.

5.2.5 BERTopic

BERTopic utilizes sentence-transformer embeddings to create document embeddings. These embeddings capture the contextual and semantic richness of text, which is then clustered to form topics. We use [BERTopic's](#)⁸ modular library to do topic modelling in this research.

Corpus Preparation:

Minimal text preprocessing is performed by removing primarily noise elements such as special characters, single letters, numbers, and PGP keys using regular expressions. This approach is used because transformer embeddings are capable of handling semantic relationships within the text.

By default, embedding generation in BERTopic is done by ("[all-MiniLM-L6-v2](#)")⁹ which is lightweight yet capable for handling semantic similarity.

Model Configuration:

Parameter	Value
Embedding Model	"all-MiniLM-L6-v2"
Dimensionality Reduction	"UMAP"
Clustering Algorithm	"HDBSCAN" (min_cluster_size=5, min_samples=3)
Vectorizer	Custom CountVectorizer applied after clustering.
Number of Topics	nr_topics=10

Table 6: Configuration for BERTopic

The number of topics is limited to 10 to maintain consistency with other topic models for comparison. The cluster size is set to a minimum of 5 documents, with a minimum sample size of 3, enabling more detailed clustering and the identification of smaller, distinct groups.

6 Evaluation & Reflection

A portion of the data overview and methodology outlined here, particularly the use of LDA for topic analysis on the Dread forum and initial results, has been previously presented in the IEEE ISSE paper under the guidance of my supervisor Dr. Vitor Jesus, where I contributed to the topic modeling analysis ([Jesus et al. 2024](#)). This dissertation extends that initial work by incorporating additional models like NMF, BERTopic and modified LDA offering a more refined approach to detecting rare but critical terms. The published paper served as an early validation of the methodology, while the results presented here demonstrate a broader application and further enhancement of these techniques to better capture dynamic patterns in Dark Web discussions.

⁸<https://maartengr.github.io/BERTopic/index.html>

⁹<https://huggingface.co/sentence-transformers/all-MiniLM-L6-v2>

The following subsections will discuss both the quantitative and qualitative analyses of the topic models discussed in the experimental setup, as well as an overview of the limitations encountered throughout the study.

6.1 Quantitative Evaluation of Topic Models

This section evaluates the performance of five topic models: Baseline LDA, LDA with Nouns and NER, Modified LDA, NMF with Nouns and NER, and BERTopic. The models are evaluated based on two primary metrics: Topic Diversity and Inverse Word Embedding Coherence (IWECC) Score. These metrics offer insights into the distinctiveness of the topics and their semantic coherence, respectively. Each model is applied yearly over 7 years, and the final scores represent weighted averages across all time periods.

The metrics, Topic Diversity and Inverse Word Embedding Coherence (IWECC) Score, were introduced in 5.2.1. Briefly, Topic Diversity evaluates the uniqueness of words across topics, while IWECC measures the semantic coherence of words in each topic.

To ensure the evaluation reflects the model performance across all years, weighted averages were calculated. This approach accounts for the varying number of posts per year, giving more weight to years with higher post volumes. For instance, if a model performed exceptionally well in a year with fewer posts, that year's influence on the overall score would be smaller compared to years with more posts. This provides a balanced and fair comparison of model performance.

Model	Weighted Avg Topic Diversity	Weighted Avg IWECC
Baseline LDA	0.6689	0.3651
LDA with Nouns and NER	<i>0.9446</i>	0.3295
Modified LDA	0.9907	0.3575
NMF with Nouns and NER	0.8426	0.4245
BERTopic	0.8478	<i>0.3677</i>

Table 7: Comparison of Topic Diversity and IWECC across models.

***Note:** Bold and highlighted values indicate the best performance, while underlined and italicized values represent the second-best performance.

Across all models, a significant distinction emerged between traditional models like Baseline LDA and more refined approaches such as Modified LDA and BERTopic. Table 7 provides the weighted average scores of topic diversity and IWECC across seven years for each model, offering a summary of the models' overall performance.

Baseline LDA exhibited the lowest topic diversity score (0.6689), indicating its difficulty in generating distinct, non-redundant topics over the years. Despite a moderate IWECC score (0.3651), the model showed limitations in identifying rare, critical terms, which are central to this research (see Table 7). Its inability to capture the domain-specific nuances highlights its inefficacy in dealing with sparse, significant terms.

In contrast, LDA with Nouns and NER significantly improved topic diversity (0.9446), reflecting the positive impact of incorporating nouns and named entities into topic generation. However, the IWECC score (0.3295) shows a slight reduction in coherence, suggesting that while topic differentiation improved, the internal word relationships weakened (see Table 7). This pattern is mirrored in Modified LDA, which also enhanced topic generation using nouns and entities. Achieving the highest topic diversity score (0.9907), Modified LDA excelled at capturing distinct terms but faced a trade-off with coherence, as indicated by its IWECC score (0.3575) (see Table 1).

The use of nouns and named entities consistently boosted diversity in these models, as seen in their superior performance compared to Baseline LDA. However, the challenge of maintaining coherence persists, especially in LDA with Nouns and NER and Modified LDA, illustrating a typical trade-off between distinctiveness and coherence when prioritizing rare terms. Refer to Figures 7, 8 and 9 to see variations over the years. It is very interesting to see topic diversity score of modified LDA stayed very stable contrasting with the baseline and noun/Entity incorporated LDA

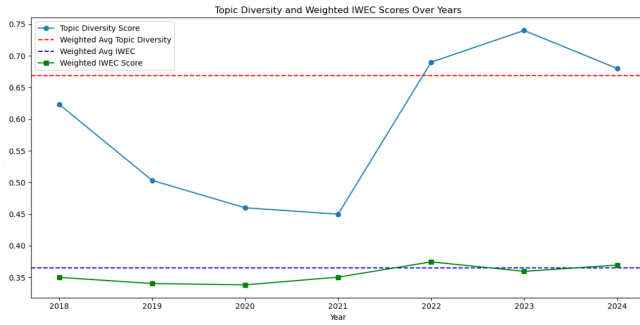


Figure 7: Baseline LDA

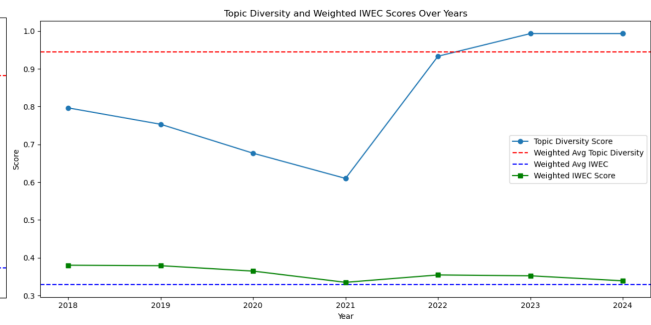


Figure 8: LDA with Nouns and NER

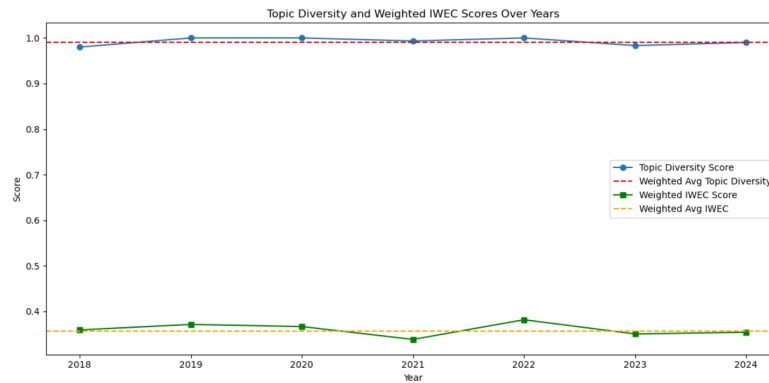


Figure 9: Modified LDA

Interestingly, NMF with Nouns and NER found a middle ground, with a high topic diversity score of 0.8426 and a stronger IWECC score of 0.4245 (refer to Table 1). By using matrix factorization, this model balanced topic diversity with coherence more effectively, providing structured topic generation with some variation from year 2021 to 2024, as reflected in Figure 10.

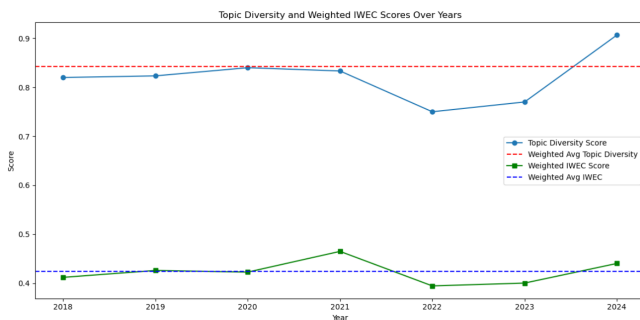


Figure 10: NMF

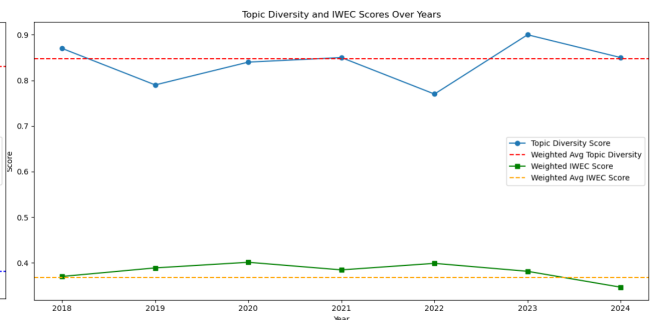


Figure 11: BERTopic

BERTopic maintained no major variation performance across both metrics, with a topic diversity score of 0.8478 and an IWECC score of 0.3677 (see Table 1). Its use of contextual embeddings enabled it to balance distinctiveness and coherence, offering stable performance.(Figure 11). Both the scores were competitive with NMF while giving the second best score in IWECC.

6.2 Qualitative Insights and Visualizations

A note on Visualizations:

For the visualizations, LDAvis introduced by Sievert & Shirley (2014) the python library pyLDAvis was used to explore and visualize the topics generated by Baseline LDA and LDA with Nouns and NER. These interactive visualizations provided insights into the structure and relationships between topics, enhancing the interpretability of the models' output. For BERTopic and NMF, word clouds were used to highlight the most frequent terms within each topic, offering a clear view of the key words that define them.

Since Modified LDA was built from scratch, there was no pre-built library available for generating visualizations, and creating one was beyond the scope of this research. As a result, no visualizations are presented for this model.

Due to space restrictions, only a glimpse of the visualizations will be provided in this section to offer a representative view of the topic structure rather than displaying the full set of visualizations across all years. As well as only important findings will be discussed.

Insights:

1. Baseline LDA

Upon observing, it was clear that baseline LDA was struggling to include contextually topic important words in its topic distribution, Topics 1, 2, and 3 across all years share common, general terms that revolve around broader discussions in the dataset. Some of the frequently occurring words include "vendor," "im," "know," "people," "like," "market," etc. indicating a focus on communications, markets, and broader issues.

Interestingly, Topic 4 displays a notable shift in focus between 2018 and 2024. In 2018, the dominant terms centered around "opioids," "fentanyl," and "brothers," highlighting discussions around drugs and illicit activities. By 2024, however, the topic has shifted significantly, with the most relevant terms now being "accounts," "bank," and "card," reflecting a stronger emphasis on finance related discussions.

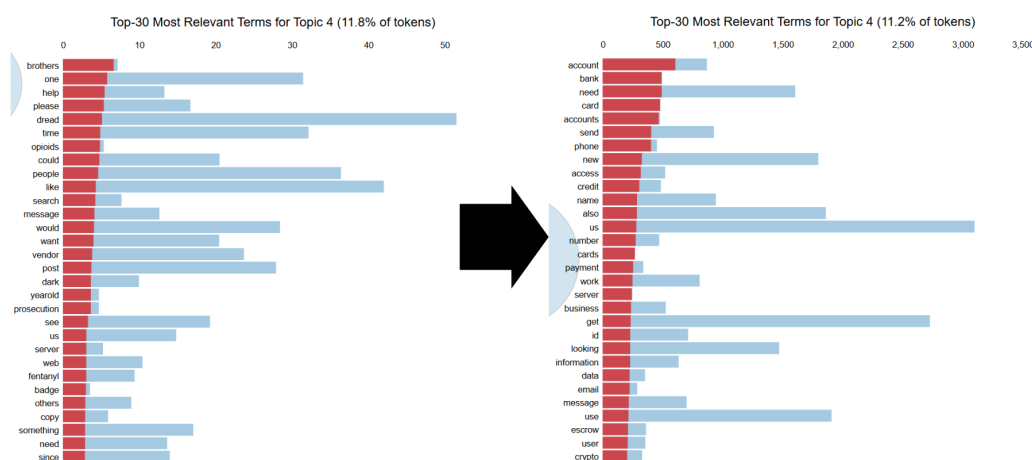


Figure 12: Topic Shift from 2018 to 2024

2. LDA with Noun and NER

By refining the LDA model to focus only on nouns and named entities (NERs), the

resulting topics became more concrete and aligned with the research objectives. The topics revolved around specific elements such as vendors, products, drug names, and geographical regions. This approach filtered out many generic terms, making the topics more actionable.

For instance, in 2024, Topic 1 includes specific market-related terms like "shipping," "ketamine," "prices," and "quality" (Figure 13), reflecting detailed discussions about the market. Topic 9 in 2024 presents intriguing terms such as "Bolivian imported," "dark-matter," and "Australian" (Figure 14), indicating niche products and vendors. However, the distribution of topics was somewhat skewed, with certain dominant themes covering large portions of the corpus.

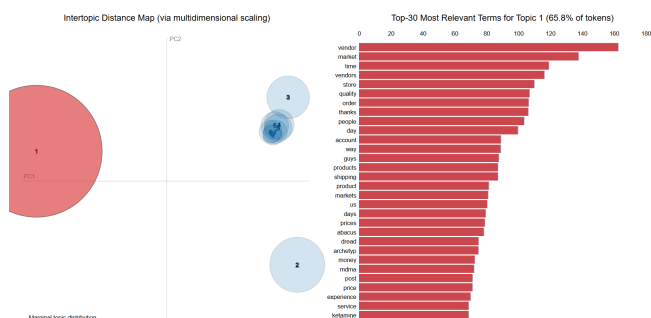


Figure 13: Topic 1 2024

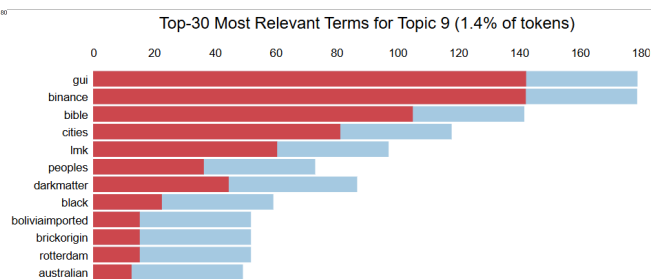


Figure 14: Topic 9 2024

3. NMF

NMF, when combined with noun and NER extraction, produced more coherent topics than LDA. While the topics still overlapped to some extent, they were more structured and informative. For example, Topic 6 in 2018 captured discussions around the "Olympus" market exit scam, with terms like "exit," "scam," "escrow," and "money" (Figure 16). Escrow refers to a third-party service that holds funds during transactions to ensure security, which is crucial in dark web markets.

Similar to LDA, NMF also revealed topic shifts, moving from general forum-related terms like "reddit" and "community" to product-specific terms such as "cocaine," "ketamine," and "purity" in later years, reflecting a shift towards product quality and shipping discussions. (see figure 15)



Figure 15: NMF capturing theme shift from 2018 to 2024



Figure 16: Topic 6 2018

4. BERTopic

In this experimental setup, BERTopic was mainly used for comparison with traditional models and not fully leveraged. Unlike LDA, BERTopic does not require specifying the number of topics in advance, which allows it to infer topics directly from the data. However, for consistency, we forced the model to reduce the number of topics to 10, which may have caused underfitting.

Despite this, BERTopic identified highly granular topics, such as discussions around Israel, which other models missed (figure 17). However, the model also included some uninformative words like "think," "maked," and "musks" in the topic representations, highlighting a potential limitation when reducing the topic count.



Figure 17: Granular topic found by BERTopic in 2024

5. Modified LDA

The modified LDA model was introduced to give more weight to rare words, aiming to enhance topic diversity, even if the words were less semantically relevant. This approach captured niche terms that were underrepresented by the other models. For example, Topic 1 in 2018 focused on Dread forum security and scams, while other topics across the years surfaced rare terms like "fraudescobay," "bitmessage" (a decentralized P2P messaging platform), "marketagartha", "cardamazon", "fullzbank", "clonesmultisig", "darkfox-market", "darknetmarketsindia", "companyamex" which were largely absent from other models' outputs.

However, this method also introduced irrelevant or typo-like words, reducing interpretability. For example, Topic 10 in 2018 included terms like "symfonycomponentdependencyinjectioncontainergetfile" and "phpbbdbdriverdriversqlerrorfile," which are PHPBB-related technical terms. Also it also features very irrelevant words typos and which may also have been rare rare but given more weight by the topic model, like "dhqkakiyyrnhehmsnorwrr" "fsjrdt". This highlights both the strengths and limitations of the modified LDA model, as it surfaces rare yet sometimes nonsensical terms.

6.3 Limitations

This study faces several key limitations that impact its depth and generalizability. A major issue is the reliance on noun and named entity (NER) extraction in models like LDA and NMF, which may overlook nuanced terms often found in the informal and cryptic language of the Dark Web. This focus on nouns and NERs can oversimplify complex discussions, missing subtle yet critical terms, especially when coded language is used.

BERTopic, while effective in using transformer-based embeddings, introduced computational challenges. Although its processing time increased due to embedding generation and clustering, the reduction to 10 topics likely limited its ability to capture the full diversity of themes in the data. In contrast, the Modified LDA, designed to emphasize rare words, was computationally expensive due to its dynamic word influence mechanism. This added complexity increased the model's burden and surfaced non-informative terms, affecting topic coherence.

The static yearly topic modeling approach also restricted the ability to track the dynamic evolution of discussions. More sophisticated models could have better captured topic shifts over time. Additionally, aggressive preprocessing, though necessary to clean the data, may have removed important low-frequency terms, further limiting insight into niche yet critical topics.

The evaluation metrics, Topic Diversity and Inverse Word Embedding Coherence (IWECC), while useful, had their shortcomings. Topic Diversity, though it measures word uniqueness, doesn't guarantee relevance, and IWECC may have overly focused on word similarity, reducing human interpretability in some cases.

Finally, the study's focus on English-language posts limits its generalizability to multilingual forums, excluding non-English discussions

7 Practical Realisation

This project applied AI methods to address critical business challenges in cybersecurity and risk management, particularly regarding Dark Web monitoring for early detection of emerging threats. Experimenting on Modified LDA model, we focused on domain specific rare and exclusive words in an unsupervised fashion which reduces the manual effort required to review large volumes of user-generated data, which is often slow and ineffective. These models, applied to Dark Web forum data, can offer businesses actionable insights to strengthen their security strategies.

By identifying rare and exclusive terms, the Modified LDA model may enable businesses to detect subtle but significant emerging threats that may go unnoticed using traditional methods. This allows organizations to act proactively, responding to potential risks like data breaches, fraud schemes, or new malware before they escalate into major security incidents. The unsupervised nature of the approach further enhances its scalability, allowing companies to continuously monitor large volumes of data without intensive human intervention.

During the analysis conducted in this project, it became clear that discussions on Dark Web forums related to drug markets and illegal substances were distinct and prevalent. Addition-

ally, frequent mentions of shipping methods, transactions and geographical locations provided valuable insights into the logistics behind these illegal activities. Such insights can be particularly useful for government organizations in identifying and tracking the distribution of illicit goods. By leveraging these AI-driven findings, authorities can gain a deeper understanding of emerging trends in illegal trade and take preemptive action to disrupt criminal networks before they proliferate.

This combination of business and governmental applications demonstrates the versatility and effectiveness of the AI models applied in this project, making them powerful tools for both risk mitigation and law enforcement.

8 Conclusion

This dissertation has explored and evaluated the application of various topic modeling techniques LDA, NMF, and BERTopic to the domain of Dark Web monitoring, with a particular focus on enhancing the detection of low-frequency but contextually significant terms. By integrating modifications, such as Named Entity Recognition (NER) and dynamic word influence in LDA, the research succeeded in highlighting critical themes that are often hidden in Dark Web communications.

The findings of the study support the hypothesis that traditional topic models, when modified, can improve the detection of significant low-frequency terms. However, despite these contributions, the research presents several limitations that must be critically acknowledged.

One key limitation was the computational complexity, especially with models like BERTopic and the modified LDA. These models, while effective in identifying important topics, required significant computational power, particularly for larger datasets. This introduces scalability issues, as applying these models to even larger Dark Web datasets or real-time monitoring systems could be problematic without access to advanced computational resources. Furthermore, the reliance on pre-trained embeddings in BERTopic introduced challenges related to memory usage and processing time, especially when working with long and complex texts.

Another important limitation was the focus on English-language datasets. Given the global nature of the Dark Web, illicit activities are often communicated in a wide range of languages. This monolingual focus limits the generalizability of the findings. Expanding this research to include multilingual datasets is crucial for a more comprehensive and accurate understanding of Dark Web trends across different regions and languages. Moreover, the study primarily relied on publicly accessible data, which, while ethically appropriate, may not fully represent the more secretive and encrypted sections of the Dark Web, potentially skewing the findings.

Additionally, the static yearly approach to topic modeling failed to capture the dynamic and fast-changing nature of discussions on the Dark Web. Although the models identified key themes for each year, this method did not account for how discussions evolve over shorter time frames, which would be vital in responding to emerging threats in real-time.

8.1 Future Work

Moving forward, there are several areas where this research can be expanded and refined. One significant avenue is the application of dynamic topic modeling. The current approach captures topics on a yearly basis, but the Dark Web is a constantly evolving space, where discussions and trends shift rapidly. A more granular approach, perhaps modeling changes over weeks or even days, would allow for the detection of emerging threats in near real-time. This could greatly benefit cybersecurity and risk management teams in a business setting by offering them more timely insights.

Another area for improvement is scalability. Some of the models used in this study, particularly BERTopic and Modified LDA, proved computationally demanding, which limits their use on larger datasets or in real-time applications. Future research could explore more efficient versions of these models or leverage distributed computing solutions to make them more practical for real-time Dark Web monitoring on a large scale.

One of the biggest challenges in monitoring Dark Web forums is the linguistic features like coded language and slang used to avoid detection. Future models could incorporate advanced techniques to interpret these changes. Large language models, such as GPT, could be fine-tuned with domain-specific datasets to improve their understanding of this evolving terminology.

Lastly, future research could benefit from better evaluation metrics. While topic diversity and IWEAC are useful, they don't fully capture the nuances of detecting important low-frequency terms in a vast amount of noise. Developing more tailored evaluation methods to assess how well models are identifying these critical terms would improve the relevance and precision of Dark Web monitoring tools.

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